

Wormhole Foundation EVM-NTT Diff Audit Summary

Prepared by Cyfrin

Version 1.0

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Wormhole Foundation on Wormhole Foundation EVM-NTT Diff. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number of findings by severity. It closes with Cyfrin's attestation of the review's scope.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 1.0 dated July 23, 2024, was delivered to Wormhole Foundation, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 4 days, the Cyfrin team conducted an audit on the Wormhole Foundation EVM-NTT Diff code provided by Wormhole Foundation. The findings consist of 4 Informational findings.

Summary

Project Name	Wormhole Foundation EVM-NTT Diff
Repository	example-native-token-transfers
Commit	0d37b0f49750...
Audit Timeline	Jul 18th - Jul 23rd, 2024
Methods	Manual Review, Stateful Fuzzing

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	0	–	–
High Risk	0	–	–
Medium Risk	0	–	–
Low Risk	0	–	–
Informational	4	–	–
Gas Optimizations	0	–	–
Total	4	–	–

This version of the report was issued before any fixes were reviewed: all findings were open.

6 Audit Scope

Cyfrin conducted a diff audit of the EVM-related Solidity contracts of the Wormhole Native Token Transfers. The previous audit was performed on commit f4e2277, and the current audit focuses on commit 0d37b0f. The scope of the audit was determined by the differences between these two commits, as outlined by the following git diff command:

```
git diff f4e2277b358349dbfb8a654d19a925628d48a8af 0d37b0f4975084492c72ca881c1218d6e1aae9e3
evm/src
```

The changes reviewed in this diff included modifications to the Solidity contracts located in the directories:

- evm/src/interfaces/*
- evm/src/NttManager/*
- evm/src/Transceiver/*
- evm/src/libraries/*
- evm/src/wormhole/*.

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

Wormhole Native Token Transfers (NTT) framework is designed to facilitate the transfer of tokens across different blockchain networks without relying on liquidity pools, offering an open, flexible and composable solution. Integrators configure NTTs on each chain, including the token standard and metadata.

9 Attestation

Cyfrin attests that it performed a security review of the Wormhole Foundation EVM-NTT Diff code provided by Wormhole Foundation.

The repositories, commits, review period and findings are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2024-07-23-cyfrin-wormhole-NTT-Diff-v1.0.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Wormhole Foundation has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=505b56fd61cc0d18ff3ce10a5929817f1be80ef3934c7713b0a742cd43289305
F=2024-07-23-cyfrin-wormhole-NTT-Diff-v1.0-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.